

Besov Spaces over Good Grids

Current Lean 4 Formalization

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Abstract

This document describes the current Lean 4 formalization in the `BesovSpacesGoodGrid` repository. The project develops a reusable weak-grid framework for atomic Besov-ish spaces, proves scale inclusions, representation-limit and compactness theorems, proves completeness for the coefficient-cost norm, and formalizes a substantial weak-grid transmutation pipeline. It also contains a good-grid specialization based on Souza atoms. The current good-grid layer also relates the standard atomic representation norm, the L^2 -normalized Haar representation norm, and the mean-oscillation norm by finite-constant comparison theorems. The principal public results are summarized in Section 11.

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1 Introduction

1.1 Purpose of the Project

The project formalizes part of the atomic decomposition theory from

Daniel Smania, *Besov-ish spaces through atomic decomposition*, Analysis & PDE 15 (2022), no. 1, 123–174.

The emphasis is on a Lean library that can be reused in later developments: the basic theory is first stated for weak grids and abstract atom families, and good-grid/Souza spaces are then obtained as a specialization.

1.2 Current Pipeline

The aggregate Lean file `BesovSpacesGoodGrid.lean` is intended to import the following pipeline:

1. define quantitative good grids and general weak grids;
2. package weak-grid cells and local Banach spaces;
3. define atom families with support, convexity, phase invariance, and atom-size estimates;
4. build levelwise atomic blocks and infinite L^p representations;
5. define the Besov-ish space and the coefficient-cost gauge `Norm_Costpq`;
6. prove scale constructions and a smoothness-scale inclusion;
7. prove representation-limit theorems, compactness statements, and completeness of the Besov-ish space for the cost norm;
8. prove transmutation claims moving atomic representations from one weak grid to another;
9. specialize the transmutation machinery to compare Souza atoms, Besov atoms, and atom families sandwiched between them on good grids;
10. introduce Haar and standard representation gauges on good grids and compare them with the mean-oscillation norm;
11. develop a pointwise-multiplier layer on good grids: the level-tail Souza `selfs` classes, boundedness of `selfs` multipliers, the $B_{1,1}^s$ multiplier characterization, and the non-Archimedean multiplier estimate.

The repository also contains good-grid Souza-space results and auxiliary block-sum notation. At the date of this document a full `lake build` succeeds; the earlier aggregate-root import/name issue involving `GoodGridSpace.GoodGridCell.toLevelCell` has been resolved.

2 Grid Structures

2.1 Good Grids

Definition 2.1 (Good grid). The Lean structure

`GoodGridSpace.GoodGrid`

extends the grid structure from the `UnbalancedHaarWavelet` dependency. It adds constants $\lambda_1, \lambda_2 : \mathbb{R}$ with

$$0 < \lambda_1 \leq \lambda_2 < 1$$

and lower and upper measure-ratio estimates for a child cell s inside a parent cell t :

$$\lambda_1 \mu(t) \leq \mu(s), \quad \mu(s) \leq \lambda_2 \mu(t).$$

Definition 2.2 (Good-grid space). The structure

`GoodGridSpace.GoodGridSpace`

is the ambient object used by good-grid files. It consists of a measurable space with a fixed good grid. The associated measure is exposed as `GoodGridSpace.measure`.

2.2 Weak Grids

Definition 2.3 (Weak grid). The Lean structure

`WeakGridSpace.WeakGrid`

is a measure together with a sequence of finite cell families $\mathcal{P}^k$. Each cell is measurable and has positive measure; the whole measure is finite; at least one level is nonempty; and there is a uniform same-level overlap bound `Cmult1`.

The term “partition” appears in field names because of the mathematical source, but a weak grid does not require disjointness or covering. The important primitive is the overlap neighborhood

$$\Omega_Q^k = \{P \in \mathcal{P}^k : P \cap Q \neq \emptyset\},$$

implemented as `overlapFinset`.

Lemma 2.1 (Overlap double counting). The theorem

`WeakGridSpace.WeakGrid.overlap_double_sum_le`

states that if f is nonnegative on the level- k cells, then summing f over all overlap neighborhoods costs at most the multiplicity constant:

$$\sum_{Q \in \mathcal{P}^k} \sum_{P \in \Omega_Q^k} f(P) \leq C_{\text{mult}1} \sum_{P \in \mathcal{P}^k} f(P).$$

3 Atoms

3.1 Cells and Local Spaces

Definition 3.1 (Weak-grid cell). The structure

`WeakGridSpace.WeakGridCell`

bundles a level k , a set Q , and the proof that $Q \in \mathcal{P}^k$. The theorem `measurable` records that every such cell is measurable.

Definition 3.2 (Local Banach space). The structure

`WeakGridSpace.LocalBanachSpace`

packages a complex normed complete vector space together with an injective linear realization map into functions $\alpha \rightarrow \mathbb{C}$. This lets different cells carry different local model spaces.

3.2 Atom Families

Definition 3.3 (Atom family). The Lean structure

`WeakGridSpace.AtomFamily`

is the central abstraction for atomic decompositions. For parameters s, p, u , it records:

- (i) the Holder conjugate u' ;
- (ii) assumptions $s > 0$, $1 \leq p < \infty$, and $1 \leq u \leq \infty$;
- (iii) a local Banach space $B(Q)$ for every weak-grid cell Q ;
- (iv) a nonempty atom set $A(Q) \subset B(Q)$;
- (v) membership of local functions in L^{pu} ;
- (vi) support inside Q ;
- (vii) convexity and phase invariance of the atom set;
- (viii) the atom-size estimate

$$\|a\|_{L^{pu}} \leq \mu(Q)^{s-1/(u'p)}.$$

The namespace `AtomFamily` exposes convenience definitions and lemmas: `toFunction`, `IsAtom`, `AtomsOn`, `allAtoms`, `atom_memLp`, `local_memLp_p`, `atom_support`, and `atom_norm_bound`. It also records optional compactness and finite-dimensional hypotheses: `StronglyCompactAtoms`, `WeaklySequentiallyCompactAtoms`, and `FiniteDimensionalAtoms`.

4 Besov-ish Spaces

4.1 Level Blocks

Definition 4.1 (Level block). For an atom family A , the structure

$$\text{WeakGridSpace.LevelBlock } A \ k$$

chooses one coefficient and one atom for each cell of the level k . Its value in L^p is the finite sum over that level:

$$\sum_{Q \in \mathcal{P}^k} c_Q a_Q.$$

Lean names this value `LevelBlock.toLp`.

The file proves that zero and scalar multiples of level blocks behave as expected. It also proves basic closure properties of the set `LevelBlockSet A k`.

4.2 Representations and Costs

Definition 4.2 (Atomic representation). The structure

$$\text{WeakGridSpace.LpGridRepresentation } A \ g$$

is a sequence of level blocks whose infinite sum is g in L^p :

$$\text{HasSum } (k \mapsto \text{toLp}(B_k)) (g).$$

Definition 4.3 (Coefficient cost). For a representation R , Lean defines `levelCoeffPower`, `lpCost`, `pqCost`, and `pqCostENNReal`. The finite-cost predicate is

$$\text{LpGridRepresentation.FinitePQCost}.$$

It is the formal version of the usual (p, q) -coefficient summability condition, with the endpoint $q = \infty$ represented by a supremum bound.

4.3 Besov-ish Space and Cost Gauge

Definition 4.4 (Besov-ish membership). The predicates

$$\text{MemBesovish}, \quad \text{MemBesovishCoeffCost}$$

say that an L^p function admits an atomic representation, respectively an atomic representation with finite (p, q) -coefficient cost.

Definition 4.5 (Besov-ish space). The Lean definition

$$\text{WeakGridSpace.BesovishSpace } A \ q$$

is the complex submodule of L^p consisting of elements satisfying `MemBesovishCoeffCost A q`.

Definition 4.6 (Coefficient-cost gauge). For $g \in \text{BesovishSpace } A \ q$, the gauge

$$\text{BesovishSpace.Norm_Costpq } A \ q \ g$$

is the infimum of the pqCost over all finite-cost representations of g .

Theorem 4.1 (Normed-space summary). The theorem

`BesovishSpace.normedSpace_and_lp_embedding`

packages the main structural facts about `Norm_Costpq`: nonnegativity, triangle inequality, homogeneity, a named cost-normed group/space structure, and an explicit L^t embedding estimate for exponents

$$p \leq t \leq pu.$$

5 Scale Inclusions

Definition 5.1 (Smoothness scaling). The file `WeakGrid/Scales.lean` defines

`smoothnessScaleFactor`, `smoothnessScaleAtoms`, `smoothnessScaleAtomFamily`.

These rescale atoms cell by cell by the deterministic factor

$$\mu(Q)^{\tilde{s}-s}.$$

Definition 5.2 (Smoothness-integrability scaling). When $u = \infty$, the file also defines the two-parameter scaling

`smoothnessIntegrabilityScaleAtomFamily`.

The scale factor is

$$\mu(Q)^{\tilde{s}-s+1/p-1/\tilde{p}}.$$

Theorem 5.1 (Smoothness-scale inclusion). The theorem

`smoothnessScaleBesovishSpace_subset`

shows that, under the deterministic summability hypothesis `scaleCoefficientFinite` and the monotonicity assumption $s \leq \tilde{s}$, the Besov-ish space for the smoothness-scaled atom family is contained in the original Besov-ish space.

Theorem 5.2 (Linear inclusion and cost bound). The definition

`smoothnessScaleBesovishSpaceInclusion`

packages the preceding set inclusion as a linear map. The theorem

`smoothnessScaleBesovishSpaceInclusion_Norm_Costpq_le`

gives the corresponding explicit bound for the cost gauge.

6 Representation Limits and Completeness

6.1 Analytic Grid Hypothesis

Definition 6.1 (Assumption G2). The predicate

`WeakGridSpace.AssumptionG2 G s p u q`

combines two requirements:

- (i) the coefficient-weight series needed for the $t = p$ embedding is finite;
- (ii) the maximal measure of cells at level k tends to zero as $k \rightarrow \infty$.

This hypothesis is used to control tails of atomic representations uniformly.

6.2 Limit Theorems

Theorem 6.1 (Representation limit). The theorem

`representation_limit`

formalizes the weak L^p version of Proposition `compa2` from the development. If representations have uniformly bounded coefficient cost, their coefficients converge cellwise, and their atoms converge weakly, then the limiting blocks define a Besov-ish function with the same cost bound and the represented functions converge weakly.

Theorem 6.2 (Strong representation limit). The theorem

`representation_limit_strong`

is the strong L^p version: strong convergence of the atoms gives strong convergence of the represented functions.

The existence theorems `representation_limit_strong_existence` and `representation_limit_w` start from a bare limit block sequence, prove it is summable, and package the resulting limit as a genuine representation.

6.3 Compactness and Completeness

Theorem 6.3 (Formal block summability). The theorem

`formalBlockSeq_summable`

says that under `AssumptionG2`, any formal block sequence with finite abstract (p, q) -cost is summable in L^p .

Theorem 6.4 (Closed cost balls). The theorem

`closed_Norm_Costpq_ball_strongly_seqCompact`

proves sequential compactness of closed coefficient-cost balls in the ambient strong L^p topology, assuming the compactness hypothesis `AssumptionA5`.

Theorem 6.5 (Completeness). The theorem

`besovishSpace_costNorm_completeSpace`

states that `BesovishSpace A q`, endowed with the local `costNormedAddCommGroup` structure associated to `Norm_Costpq`, is complete.

7 Weak-Grid Transmutation

7.1 Setup

The file `WeakGrid/Transmutation.lean` formalizes the transfer of an atomic representation from a source weak grid G to a target weak grid W . The source coefficients are measured by `CoeffPLevel`, `CoeffPQCost`, and `CoeffFinitePQCost`.

Definition 7.1 (Almost linear level selector). The predicate

`AlmostLinearSequence k`

says that the target level $k(i)$ attached to a source level i stays between two affine functions with the same positive slope:

$$ri + A \leq k(i) \leq ri + B.$$

It is used to prove finiteness of certain level interactions and to reduce coefficient estimates to convolution bounds.

Definition 7.2 (Representation hypothesis). The predicate

`RepresentationWsubGandALS`

records the structural hypothesis that every source atom has a target-grid atomic representation with localization, decay, and almost-linear level control.

7.2 Transmutation Objects

For each truncation level N , the file defines:

- `PartialSumLevels`: the source partial expansion;
- `TransmutationCoeff`: the target coefficient $m_{P,N}$;
- `TransmutationAtom` and `TransmutationAtomLocal`: the normalized target atoms $d_{P,N}$;
- `TransmutationBlock`: the target level block at level j .

It also defines stable limiting objects: `TransmutationCoeffLimit`, `TransmutationAtomLimit`, and `TransmutationBlockLimit`.

7.3 Claims

Theorem 7.1 (Claim I). The theorem

`ClaimI`

is the exact bookkeeping identity. The transmuted target representation at truncation N sums to the source partial expansion `PartialSumLevels ... N`. The theorem `ClaimI_top` is the endpoint $q = \infty$ version.

Theorem 7.2 (Claim II). The theorem

ClaimII

states that each finite transmutation block sequence is a genuine Besov-ish representation of the source partial expansion and that its (p, q) -cost is bounded by the source coefficient cost times explicit geometric, decay, and convolution constants. The theorem `ClaimII_top` treats the endpoint $q = \infty$.

Theorem 7.3 (Claim III). The theorem

ClaimIII

passes to the stable limiting transmutation data. It produces an L^p -limit g_∞ , proves that the stable blocks form a Besov-ish representation of g_∞ , and proves strong convergence of the truncated source sums to that limit. The theorem `ClaimIII_top` is the endpoint version.

Theorem 7.4 (Identity-level Claim C constant). For the identity level selector used in the good-grid Besov-atom comparison, the theorem

`LpGridRepresentation.cCoefficientInt_transmutationKernelZ_zero_one`

computes the integer transmutation coefficient of the Claim C kernel as the ordinary geometric sum:

$$C_{co, \mathbb{Z}}(p, \infty, n \mapsto \lambda^n \mathbf{1}_{n \geq 0}) = (1 - \rho)^{-1}, \quad \lambda = \rho^p.$$

This is the constant simplification used in the final Souza/Besov atom comparison.

8 Good-Grid Souza Spaces

The file `GoodGrid/BesovSpace.lean` specializes the weak-grid theory to good grids and Souza atoms.

Definition 8.1 (Induced weak grid). The definition

`GoodGridSpace.toWeakGrid`

turns a good grid into a weak grid. Since good-grid partitions are disjoint, the same-level overlap multiplicity is 1. The induced weak-grid space is `GoodGridSpace.toWeakGridSpace`.

Definition 8.2 (Souza atoms). The predicate

`GoodGridSpace.IsSouzaAtom`

says that an atom is supported on a good-grid cell, is constant on that cell, and has size at most

$$\mu(Q)^{s-1/p}.$$

The definitions `canonicalSouzaAtom`, `souzaLocalBanachSpace`, `souzaAtomsSet`, and `souzaAtomFamily` package these atoms as an `AtomFamily` with $u = \infty$.

Definition 8.3 (Souza Besov space). The definition

`GoodGridSpace.SouzaBesovSpace`

is the Besov-ish space associated to the Souza atom family on the weak grid induced by the good grid.

Theorem 8.1 (Good-grid completeness). The theorem

`souzaBesovSpace_costNorm_completeSpace`

specializes the weak-grid completeness theorem to Souza atoms on a good grid.

Theorem 8.2 (Compact balls). Theorems

`souza_closedCostBallInLp_isSeqCompact`,
`souza_closedCostBallInLp_isCompact`, `souza_closedCostBallInL1_isCompact`

state compactness of closed cost balls in the ambient L^p topology and, after the finite-measure inclusion, in L^1 .

Theorem 8.3 (Density). Theorems

`souzaBesovSpace_dense`, `souzaBesovSpace_dense_inL1`

prove density of the Souza Besov space in the ambient L^p space and density of its image in L^1 .

8.1 Besov Atoms and the Souza/Besov Comparison

The file `GoodGrid/BesovAtoms.lean` introduces Besov atoms on a good grid. A Besov atom on a cell Q is a function supported on Q whose induced Besov-ish norm of order β is controlled by the scale $\mu(Q)^{s-\beta}$. The construction is phrased using induced weak grids and the transmutation API.

Definition 8.4 (Besov-atom normalizing constant). The definition

`besovAtomConstant`

is the geometric normalizing constant for Besov atoms. The theorem `besovAtomConstant_nonneg` proves nonnegativity, and `one_le_besovAtomConstant` proves the lower bound $1 \leq \text{besovAtomConstant}$, which is used to simplify the final Claim C constant.

Theorem 8.4 (Besov atom to Souza representation). The theorem

`besovAtom_to_induced_souzaS_representation_decay_claimC`

packages the paper's transmutation claim for a single Besov atom: after passing to the weak grid induced on its supporting cell, the atom admits a Souza-atom representation with geometric coefficient decay

$$(\lambda_2^{\beta-s})^{pk}.$$

Theorem 8.5 (Souza atoms and Besov atoms). The theorem

`atoms_between_souza_atoms_and_besov_atoms`

formalizes the sandwich proposition. If an atom family A lies between Souza atoms and Besov atoms, then the Besov-ish spaces generated by Souza atoms, by A , and by Besov atoms are equal. The theorem also gives the continuous norm bounds

$$\|f\|_A \leq C_1 \|f\|_{\text{Souza}}, \quad \|f\|_{\text{Souza}} \leq \frac{C_2}{1 - \lambda_2^{\beta-s}} \|f\|_A.$$

The second inequality is obtained by applying explicit Claim C and the geometric-sum constant computation above.

8.2 Haar, Standard, and Mean-Oscillation Norms

The newer good-grid files formalize three analytic gauges associated with an integrable function f :

- the standard atomic representation norm `standardRepresentationNorm`;
- the L^2 -normalized Haar representation norm `haarL2RepresentationNorm`;
- the mean-oscillation norm `meanOscillationNorm`, built from a global L^p term and levelwise local oscillations.

Theorem 8.6 (Haar controls standard). The theorem

`exists_standardRepresentationNorm_le_const_mul_haarL2RepresentationNorm`

states that, for fixed good-grid data and parameters, the standard atomic representation norm is bounded by a finite constant times the Haar representation norm.

Theorem 8.7 (Finite standard norm endpoint). The theorem

`finite_standardRepresentationNorm_implies_memBesov_and_standardRepresentation`

packages the endpoint consequence of finite standard norm. It gives L^p membership, the canonical standard `LpGridRepresentation`, finite (p, q) -coefficient cost, and membership in the Souza Besov space, with the expected cost bounds.

Theorem 8.8 (Finite Haar norm endpoint). The theorem

`finite_haarL2RepresentationNorm_implies_memLp_and_hasSum`

states that finite Haar representation norm implies $f \in L^p$ and that the normalized Haar expansion has sum f in L^p .

Theorem 8.9 (Standard controls mean oscillation). The theorem

`exists_meanOscillationNorm_le_const_mul_standardRepresentationNorm`

proves the manuscript direction in which the standard representation norm controls the mean-oscillation norm. The proof separates the L^p term from the oscillation seminorm. The oscillation estimate uses the fact that low-frequency Souza blocks are constant on finer cells, so local oscillation is controlled by the high-frequency tail, followed by a geometric discrete convolution estimate.

Theorem 8.10 (Mean oscillation controls Haar). The theorem

`exists_haarL2RepresentationNorm_le_const_mul_meanOscillationNorm`

proves the reverse analytic comparison from mean oscillation to Haar norm. The cellwise estimate is the Lean analogue of testing $f - c_Q$ against the zero-mean Haar wavelets over a parent cell Q :

$$|d_S| \leq C \operatorname{osc}_p(f, Q) \mu(Q)^{1/p' - 1/2}.$$

After summing over branches and levels, this gives a finite constant C such that

$$\|f\|_{\text{Haar}} \leq C \|f\|_{\text{mean osc}}.$$

9 Pointwise Multipliers

The files in `GoodGrid/Multipliers/` formalize the pointwise-multiplier theory of Souza Besov spaces, following the multiplier sections of the paper.

9.1 The selfs classes

A function g is tested as a multiplier through its products with the Souza atoms. For a tail level t , `SouzaPointwiseSelfsTailBound` $G \text{ b } p \text{ q } t \text{ g } C$ records that, for every Souza atom on a cell of level at least t , the product $g \cdot a_Q$ admits a $B_{p,\tilde{q}}^\beta$ -representation of cost at most C ; `SouzaPointwiseSelfsTailClass` is the qualitative “ g has some finite such bound”, and `souzaPointwiseSelfsTailNorm` is the induced $\mathcal{B}_{p,\tilde{q},\text{selfs}}^{\beta,t}$ seminorm. These definitions live in `Multipliers/Definition.lean`.

9.2 Boundedness

The file `Multipliers/MultipliersareBounded.lean` proves that `selfs` multipliers are bounded. Starting from local L^p/L^1 and integral estimates with the same constant as the global theory, the main outputs are

$$\|g(x)\| \leq C(G, s, p, q) |g|_{\mathcal{B}_{p,\tilde{q},\text{selfs}}^{\beta,t}} \quad \text{for a.e. } x,$$

namely `souzaPointwiseSelfsTailBound_norm_ae_le` and `souzaPointwiseSelfsTailNorm_norm_ae` with a constant independent of the tail level t .

9.3 The $B_{1,1}^s$ characterization

In `Multipliers/Besovs11.lean`, the theorem `souzaPointwiseMultiplier_iff_souzaPointwiseSelfs` identifies, for $p = q = 1$, the pointwise multipliers of $B_{1,1}^s$ with the `selfs` class.

9.4 Non-Archimedean behaviour

The file `Multipliers/NonArchimedeanProperty.lean` proves the non-Archimedean estimate. Given a family of `selfs` multipliers g_i , $i \in \Lambda$, and $f = \sum_k \sum_Q c_Q a_Q$ satisfying a bounded overlap hypothesis (the total `selfs` seminorm of the multipliers active on any cell with $c_Q \neq 0$ is at most N) and a level/scale hypothesis, the product $(\sum_i g_i) f$ has a representation whose coefficient cost is at most $C_{\text{gen}} N$ times that of f — with C_{gen} independent of the multipliers, of N , and of Λ . This is `souzaNonArchimedeanPropertyLambdaFinite` (finite Λ) and its infinite-index pointwise refinement `souzaNonArchimedeanProperty`.

9.5 Positive cone (in progress)

A positive-cone version, `souzaNonArchimedeanPropertyPositiveCone`, is being formalized. Here only the source representation’s atoms are assumed canonical (coefficients are arbitrary complex numbers), and the two consequences of the paper’s positivity remark are separated by their true strengths: the *support* consequence (every active output cell lies in the support of some multiplier) is unconditional, while the *positivity* consequence holds only when the source coefficients are nonnegative and is stated with the weaker `SouzaConePositiveRepresentation` predicate (nonnegative coefficients and atoms in

the positive cone, rather than canonical atoms). The statement, the assembly core, the with-errors layer (representation, cost, support, and cone-positivity), and the $N > 0$ clean-constant layer are proved; two **sorrys** remain (the $N = 0$ degenerate case and the positive local transmutation-data builder).

10 Auxiliary Files

10.1 Distributions

The file `GoodGrid/Distribution.lean` defines:

- `atomIndicator`: a simple-function indicator attached to a good-grid cell;
- `atomIndicators`: the set of all such indicators;
- `TestFunctions`: the real submodule spanned by the indicators;
- `Distributions`: the algebraic dual of the test-function module.

10.2 Block-Sum Notation

The file `Sums.lean` provides reusable indexing types and notation for block sums:

- `BlockIndex` for global pairs (k, Q) ;
- `ChildIndex` and `ParentIndex` for strict descendants and ancestors;
- `blockTsum`, `childTsum`, and `parentTsum`;
- notation such as `sumb`, `sum+`, and `sum-` in the source file.

10.3 Repository Status

At this snapshot, a full `lake build` succeeds: the aggregate root file `BesovSpacesGoodGrid.lean` and all imported modules compile. The earlier import/name issue around `GoodGridSpace.GoodGridCell` has been resolved. Project Lean files outside `.lake/packages` contain no `admit` and no project-local `axiom` or `constant` declarations. The only Lean `sorry`s are the two in the in-progress positive-cone multiplier file `GoodGrid/Multipliers/NonArchimedeanProperty.lean`; they build as `sorry` warnings and do not block the project build. The current verification log is kept in `status.md`.

11 Main Theorems

The most important public results currently include:

- `BesovishSpace.normedSpace_and_lp_embedding`: structural summary for the coefficient-cost gauge and the L^t embedding estimate.
- `smoothnessScaleBesovishSpace_subset` and `smoothnessScaleBesovishSpaceInclusion_Norm`: inclusion and norm bound for smoothness-scaled Besov-ish spaces.
- `representation_limit` and `representation_limit_strong`: weak and strong representation-limit theorems.
- `representation_limit_weak_existence` and `representation_limit_strong_existence`: existence-packaged forms of the representation-limit statements.
- `closed_Norm_Costpq_ball_strongly_seqCompact`: sequential compactness of closed cost balls in the strong L^p topology.
- `besovishSpace_costNorm_completeSpace`: completeness of the Besov-ish space for `Norm_Costpq`.
- `ClaimI`, `ClaimII`, and `ClaimIII`, plus the `_top` endpoint versions: the weak-grid transmutation pipeline.
- `Transmutation_of_Atoms_continuous_embedding_explicit` and `LpGridRepresentation.cCo`: explicit Claim C embedding and the identity-level geometric constant.
- `atoms_between_souza_atoms_and_besov_atoms`: equality of the Souza, intermediate, and Besov-atom Besov-ish spaces with the paper's norm constants.
- `souzaBesovSpace_costNorm_completeSpace`, `souza_closedCostBallInLp_isCompact`, `souza_closedCostBallInL1_isCompact`, `souzaBesovSpace_dense`, and `souzaBesovSpace_dense_inL1`: good-grid/Souza consequences.
- `exists_standardRepresentationNorm_le_const_mul_haarL2RepresentationNorm`: the Haar representation norm controls the standard atomic representation norm.
- `finite_standardRepresentationNorm_implies_memBesov_and_standardRepresentation`: finite standard norm gives L^p , a canonical representation, finite cost, and Souza-Besov membership.
- `finite_haarL2RepresentationNorm_implies_memLp_and_hasSum`: finite Haar norm gives L^p membership and convergence of the normalized Haar expansion to f .
- `exists_meanOscillationNorm_le_const_mul_standardRepresentationNorm`: standard representation norm controls the mean-oscillation norm.
- `exists_haarL2RepresentationNorm_le_const_mul_meanOscillationNorm`: mean oscillation controls the Haar representation norm.
- `souzaPointwiseSelfsTailBound_norm_ae_le` and `souzaPointwiseSelfsTailNorm_norm_ae_le`: tail-selfs multipliers are bounded, with an L^∞ estimate uniform in the tail level.

- `souzaPointwiseMultiplier_iff_souzaPointwiseSelfsClass_one_one`: the $B_{1,1}^s$ pointwise-multiplier characterization.
- `souzaNonArchimedeanPropertyLambdaFinite` and `souzaNonArchimedeanProperty`: the finite and infinite-index non-Archimedean multiplier estimates.
- `souzaNonArchimedeanPropertyPositiveCone` (in progress): the positive-cone version, with the support and cone-positivity consequences separated; see Section 9.5.

12 File Guide

- `BesovSpacesGoodGrid.lean`: main library entry point.
- `GoodGrid/Definition.lean`: good-grid structures.
- `WeakGrid/Definition.lean`: weak-grid structures and overlap estimates.
- `WeakGrid/Atoms.lean`: cells, local spaces, and atom families.
- `WeakGrid/BesovishSpaces.lean`: level blocks, representations, costs, Besov-ish spaces, and the cost gauge.
- `WeakGrid/Scales.lean`: scaled atom families and inclusions.
- `WeakGrid/Completeness.lean`: limit, compactness, and completeness theorems.
- `WeakGrid/InducedGrid.lean`: induced weak grids on a fixed cell and the contraction into the ambient Besov-ish space.
- `WeakGrid/Transmutation.lean`: transmutation objects and Claims I, II, III, including the explicit Claim C embedding constants.
- `GoodGrid/BesovSpace.lean`: Souza atoms and good-grid Besov spaces.
- `GoodGrid/BesovAtoms.lean`: Besov atoms on good grids and the Souza/Besov atom comparison theorem.
- `GoodGrid/Multipliers.lean` and `GoodGrid/Multipliers/`: the pointwise-multiplier layer — `Definition.lean` (the `selfs` classes and seminorm), `Besovspq.lean` (induced-cell representations and restriction infrastructure), `Besovs11.lean` (the $B_{1,1}^s$ characterization), `MultipliersareBounded.lean` (boundedness of `selfs` multipliers), `NonArchimedeanProperty.lean` (the non-Archimedean estimate and the in-progress positive-cone version), and `NonArchimedeanPropertyPositiveStandalone.lean` (the user-facing positive-cone statement).
- `GoodGrid/AlternativeRepresentationsAndNorms.lean`: public aggregator for the Haar, standard, and mean-oscillation representation and norm-comparison files.
- `GoodGrid/AlternativeRepresentationsAndNorms/HaarRepresentationNorm.lean`: normalized Haar coefficients and the Haar representation gauge.

- `GoodGrid/AlternativeRepresentationsAndNorms/standardRepresentation.lean`: standard atomic coefficients and the standard representation gauge.
- `GoodGrid/AlternativeRepresentationsAndNorms/standardNormleqHaarRepresentstionNorm`: control of the standard representation norm by the Haar representation norm.
- `GoodGrid/AlternativeRepresentationsAndNorms/FiniteStandardNormimpliesBesov.lean`: finite standard norm endpoint and Souza-Besov membership.
- `GoodGrid/AlternativeRepresentationsAndNorms/FiniteHaarNormimpliesLp.lean`: finite Haar norm endpoint.
- `GoodGrid/AlternativeRepresentationsAndNorms/MeanOscillationNorm.lean`: mean-oscillation definitions and reusable local oscillation lemmas.
- `GoodGrid/AlternativeRepresentationsAndNorms/OscillationNormleqBesovNorm.lean`: control of mean oscillation by the standard representation norm.
- `GoodGrid/AlternativeRepresentationsAndNorms/HaarNormleqOscillationNorm.lean`: control of the Haar representation norm by mean oscillation.
- `GoodGrid/Distribution.lean`: test functions and distributions.
- `Sums.lean`: block-index and block-sum infrastructure.

13 Conclusion

The repository now contains a substantial weak-grid atomic-decomposition library: it defines Besov-ish spaces, develops their coefficient-cost norm, proves scale, compactness, completeness, and transmutation statements, and specializes the theory to Souza atoms on good grids. The good-grid layer now also contains the main finite-constant comparisons among the standard atomic norm, Haar representation norm, and mean-oscillation norm, together with a pointwise-multiplier layer: the `selfs` classes, boundedness of `selfs` multipliers, the $B_{1,1}^s$ characterization, and the non-Archimedean multiplier estimate, with a positive-cone version in progress. The most reusable API is currently the weak-grid layer; the good-grid files show how the abstract machinery is intended to specialize to concrete Besov spaces and to their pointwise multipliers.